

RESUME

a. Personal Information			
Name: Dr. Sammer Zai			
Father's Name: Rasheed Khan Yousuf Zai			
1.	i. Field of Specialization: Computer Science & Engineering ii. Research Interests: Image Processing iii. Expertise: Data Structures & Algorithm Analysis, Computer Vision, Artificial Intelligence, Data Visualization		
2.	Designation with Grade and department / institute: Associate Professor (BPS-20), Department of Computer Systems Engineering, Mehran University of Engineering & Technology, Jamshoro.	3.	Administrative Post (Additional) NA
4.	Domicile: Hyderabad (Urban)	5.	Date of Birth: 04.03.1987
6.	Religion: Islam	7.	Gender: Female
8.	PEC / PCATP Registration No.: COMP/06008	9.	Passport No.: CU 6910712
10.	Date of Joining of University: 24.03.2009	11.	Status of PhD Supervision: NA
12.	Tel.: Off.: 022-2772250-60 (4208) Fax: Res.: - Cell: 0334-2618325	13.	Email Address: Official: sammer.zai@faculty.muett.edu.pk sammer.zai@teacher.muett.edu.pk
14.	Marital Status: Married	15.	CNIC No.: 41304-2876071-2
16. HEC ENGAGEMENTS NA			
Research / Project Students Under supervision			
Undergraduate		Masters	
Several groups		04(completed) 08 (In progress)	
PhD		-	

b. Education (Please start from Graduation Degree onwards)				
Date		Name & Address of Institution	Academic Degree / Diploma / Certificate Obtained	Subject / Field of Study
From	To			
2004	2009	Mehran University of Engineering & Technology, Jamshoro	Bachelor of Engineering	Computer Systems Engineering
2009	2012	Mehran University of Engineering & Technology, Jamshoro	Master of Engineering	Communication Systems and Networks

2014	2017	Hanyang University, Ansan, South Korea	PhD	Computer Science and Engineering
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c(i). Academic Experience (Please start with the most recent experience) *				
Designation / Title	Date		Name of Organization where Working or Worked	Brief Job Description
	From	To		
ASSOCIATE PROFESSOR	10 Jan 2022	To Date	Department of Computer Systems Engineering, MUET, Jamshoro	Research & Teaching
ASSISTANT PROFESSOR	11 Sep 2017	09 Jan 2022	Department of Computer Systems Engineering, MUET, Jamshoro	Research & Teaching
LECTURER (REGULAR)	03 Mar 2009	10 Sep 2017	Department of Computer Systems Engineering, MUET, Jamshoro.	Research & Teaching

* Please add extra rows, if more space is required.

Research Interests	Subjects Taught
- Digital Image Processing ✓ Segmentation of Coronial Arterial tree from CTA data ✓ Detection of Coronary Plaque ✓ Seed detection for active contour methods ✓ Plant disease detection - Image Analysis - Machine Learning	✓ Introduction to computer programming ✓ Data structure and algorithm analysis ✓ Object oriented programming ✓ Operating systems concepts ✓ Data mining ✓ Data Visualization ✓ Artificial Intelligence

Honors & Rewards
- Received Certificate of Appreciation for being Session Chair during 2nd International Conference on Computational Sciences and Technologies (INCCST'20) held from 17th to 19th December 2020. - Received Certificate of Appreciation for successfully organizing the 2nd International Conference on Computational Sciences and Technologies (INCCST'20) held from 17th to 19th December 2020. - Scholarship awarded in March 2014 by Higher Education Commission (HEC) for pursuing PhD degree from South Korea. - A shield of honor has been awarded for the contribution and services in SCONEST 2011(Student Conference on Engineering Sciences & Technology) Conference on Engineering Sciences & Technology) held on 27th-28th July 2011 organized by Mehran, UET Jamshoro in collaboration with Higher Education Commission, IEEE Inc, USA Karachi Section. - A shield of honorable Judge has been awarded for the judgment of Speed Programming competition in National Engineering eXhibition'11(NEX'11), held on 2nd November 2011 at Faculty of Electrical, Electronics and Computer Engineering (FEECE), Mehran UET, Jamshoro.

- A Certificate of appreciation for dedicated efforts and professional demeanor in organizing SOFTjinzhang'11 competition held on 5th October 2011 organized by Department of Computer Systems Engineering at Mehran University of Engineering Technology, Jamshoro.
- Received a Gold Medal for being the Best Graduate of 05-Batch in B.E (Department of Computer System Engineering) from Mehran UET (2009).
- Received a Silver Medal and a Certificate of Merit for securing 1st position in B.E (Department of Computer System Engineering) from Mehran, UET (2009).
- Received a Gold Medal and a Certificate of Merit for securing 1st position in intermediate from B.I.S.E Hyderabad (2004), Pakistan.

d. Other Activities		
Sr. No.	Description	Number
i.	Total Number of Research Papers Published in National Journals . Please give details in Table No. 1	4
ii.	Total Number of Research Papers Published in International Journals . Please give details in Table No. 2	17
iii.	Total Number of Research Papers Presented at Conferences in Pakistan . Please give details in Table No. 3	5
iv.	Total Number of Research Papers Presented at Conferences outside Pakistan . Please give details in Table No. 4	1
v.	Total Number of Conferences / Seminars / Symposia / Workshops attended in Pakistan . Please give details in Table No. 5	9
vi.	Total Number of Conferences / Seminars / Symposia / Workshops attended outside Pakistan . Please give details in Table No. 6	2
vii.	Total Number of Memberships of National Professional Bodies / Councils / Committees / Organizations	1

* Please add extra rows, if more space is required.

Table No. 1

Detail of Research Papers Published in National Journals*:

Sr. No.	Title of Research Paper	Date of Publication	Complete Name and Address of the Journal	HEC Category (W, X, Y, Z) / Impact Factor	Volume Number	Issue Number	Page(s) Number	
							From	To
1.	Double Channel CNN Based Tomato Plant Leaf Disease Detection	2021	Quaid-e-Awam University Research Journal of Engineering	Y	19	2	1	7
2.	Privacy preserving location-based services through k-anonymized vehicular social network	2020	Quaid-e-Awam University Research Journal of Engineering	Y	18	2	163	168

3.	Performance comparison of RFID and barcode technologies for library management system	2018	Engineering Science and Technology International Research Journal	-	2	3	8	15
4.	Performance analysis of TCP over UMTS network	2012	Mehran University Research Journal of Engineering & Technology	-	32	4	551	558

* Please add extra rows, if more space is required.

Table No. 2

Detail of Research Papers Published in International Journals*:

Sr. No.	Title of Research Paper	Date of Publication	Complete Name and Address of the Journal	Thomson ISI Indexed / Indexing / Abstracting Service / Impact Factor	Volume Number	Issue Number	Page(s) Number	
							From	To
1.	Compact Convolutional Neural network architecture for Onion disease classification using crop images	10-01-2025	Advances and Applications in Discrete Mathematics	HEC Indexed (X category)	42	3	253	271
2.	Deep learning based multi crop disease detection system	July 2024	International Journal of Innovations in Science & Technology	HEC Indexed (Y category)	6	3	1009	1020
3.	Deep learning-based identification and categorization of various phases of diabetic retinopathy	June 2024	International Journal of Innovations in Science & Technology	HEC Indexed (Y category)	6	2	772	784
4.	Investigating deep learning methods for detecting Lung Adenocarcinoma with the TCIA dataset	2023	International Journal of Innovations in Science & Technology	HEC Indexed (Y category)	5	4	746	759
5.	Comparative analysis of TF-IDF and hashing vectorization for fake news detection in Sindhi: A machine learning and deep learning approach	2023	Engineering Proceedings	HEC Indexed (Y category)	46	5	1	7

6.	Onion crop monitoring with multispectral imagery using deep neural network	2021	International Journal of Advanced Computer Science and Applications	Scopus	12	5	303	309
7.	Image based Onion Disease (Purple Blotch) detection using deep convolutional network	2021	International Journal of Advanced Computer Science and Applications	Scopus + Web of Science	12	5	448	458
8.	Isolation of coronary arteries based on active contour and region growing models with automated seed selection	2021	Journal of Theoretical and Applied Information Technology	Scopus	99	7	1669	1678
9.	Analysis of swarm intelligence routing protocols in large-scale mobile ad hoc networks (MANET's)	2021	Journal of Theoretical and Applied Information Technology	Scopus	99	2	269	278
10.	Detection of NCOV-19 from hybrid dataset of CXR images using deep convolutional neural network	2020	International Journal of Advanced Computer Science and Applications	Scopus	11	12	699	707
11.	Development of an android app for text detection	2019	Journal of Theoretical and Applied Information Technology	Scopus	97	20	2485	2496
12.	Robust seed detection for coronary arteries segmentation using thresholded Frangi response	July 2017	Turk J Elec Eng & Comp Sci	SCIE/0.682	25	-	2749	2760
13.	Automatic and effective delineation of coronary arteries from CTA data using two-way active contour model	April 2017	IEICE TRANS. INF. & SYST.	SCIE/0.576	E100-D	4	901	909
14.	Segmentation of coronary arterial tree using localized deformable model embedded with automated seeds	April 2017	Journal of Theoretical and Applied Information Technology	Scopus	95	7	1565	1572
15.	Analysis of soft plaque detection methods in CTA images	Mar 2017	Journal of Theoretical and Applied Information Technology	Scopus	95	6	1410	1417

16.	Automatic segmentation of coronary arteries from computed tomography angiography data cloud using optimal thresholding	Jan 2017	Optical Engineering	SCI/1.082	56	1	01310 6-1	0131 06-8
17.	Performance Comparison of Vesselness Measures for Segmentation of Coronary Arteries in 2D Angiograms	Dec 2016	Indian Journal of Science and Technology	Scopus (2013 – 2016)	9	48	1	8

Table No. 3

Detail of Research Papers Presented at Conferences in Pakistan*:

Sr. No.	Title of Research Paper	Title of Conference	Venue	Date of Conference	Duration
1.	Bimodal Emotion Recognition: A Deep Multimodal Approach Using Audio and Visual Signals	1st International Conference on Innovations in Information and Communication Technologies (IICT)	MUET Jamshoro	15 th – 17 th January 2026	3 days
2.	Fault Detection in Solar Plates Using Deep Learning	8th International MultiTopic ICT Conference (IMTIC),	MUET Jamshoro	29 th - 31 st December 2025	3 days
3.	An efficient approach towards delineation of lungs and pulmonary nodule detection	2nd International Conference on Computational Sciences and Technologies (INCCST'20)	MUET Jamshoro	17 th – 19 th December 2020	3 days
4.	Performance comparison of machine learning classifiers for the detection of potato leaf diseases	2nd International Conference on Computational Sciences and Technologies (INCCST'20)	MUET Jamshoro	17 th – 19 th December 2020	3 days
5.	Improving vascular segmentation by geometric fitting in level set evolution	IEEE International Conference on Innovations in Electrical Engineering and Computational Technologies (ICIEECT)	Indus University, Jamshoro	5 th – 7 th April 2017	3 days

* Please add extra rows, if more space is required.

Table No. 4**Detail of Research Papers Presented at Conferences outside Pakistan*:**

Sr. No.	Title of Research Paper	Title of Conference	Venue and Host Country	Date of Conference	Duration
1.	An effective enhancement and segmentation of coronary arteries in 2d angiograms	IEEE, International Conference on Smart Computing and Electronic Enterprise, ICSCEE	Kuala Lumpur, Malaysia	11 th – 12 th July 2018	2 days

* Please add extra rows, if more space is required.

Table No. 5**Details of Conferences / Seminars / Symposia / Workshops attended in Pakistan*:**

Sr. No.	Name of Event	Type of Event	Chief Guest / Guest Speaker	Organizing Agency	Venue	Date(s) of Event
1.	OBE Implementation training for CS faculty	Workshop	Dr. Faisal Khan (Hamdard University) Prof. Dr. Awais Khatri	CSE, MUET	MUET	6 th & 12 th October 2022
2.	Outcome based education and assessment system of Accreditation (OBE/OBA)	Training	Dr. Naeem Mahoto	SWE, MUET	MUET	19 th April 2022
3.	Capacity Building Workshop on Developing SAR as per PEC Accreditation Manual 2014 for OBE&A Implementation	Workshop	-	PEC / MUET	MUET	05 December 2017
4.	LRQA Internal QMS Auditor Course based on ISO 9001-2008	Training	-	MUET	MUET	22 nd – 23 rd August 2013
5.	Academic and Technical Writing Skills	Workshop	-	Al Bar Academy	MUET	2 nd May 2009
6.	MATLAB Programming with Hardware Interfacing	Workshop / Training	-	Rasta Educational Network	MUET	23 rd – 24 th August 2008
7.	Educational Training on Lab Equipment	Training	-	National Scientific Cooperation	MUET	10 th June – 10 th July 2008
8.	Creating Social Semantics Web 2.0 Information Space	Workshop	-	IMTIC'08	MUET	12 th April 2008
9.	Instrumentation Interface Design and Automation in Research and Industry	Workshop	-	JCI	MUET	10 th September 2007

* Please add extra rows, if more space is required.

Table No. 6

Details of Conferences / Seminars / Symposia / Workshops attended outside Pakistan*:

Sr. No.	Name of Event	Type of Event	Chief Guest / Guest Speaker	Organizing Agency	Venue and Country	Date(s) of Event
1.	CENTRAL International Workshop on MOODLE based Interactive Content Development	Workshop	-	Kolhapur Institute of Technology, India & Technical University of Sofia, Bulgaria	Online	27 th – 30 th July 2020
2.	IEEE International Conference on Smart Computing and Electronic Enterprise	Conference	-	IEEE	Kuala Lumpur, Malaysia	11 th – 12 th July 2018

Table No. 7

Details of Memberships of National Professional Bodies / Councils / Committees / Organizations*:

Sr. No.	Name of Bodies / Councils / Committees / Organizations	Date and Duration of Membership
1.	Pakistan Engineering Council	2009 - Life time

* Please add extra rows, if more space is required.